

360카메라, 3D Gaussian Splatting을 이용한 시설물 3D구현

seminar

[2024.05.02]
AI 연구센터
이동근

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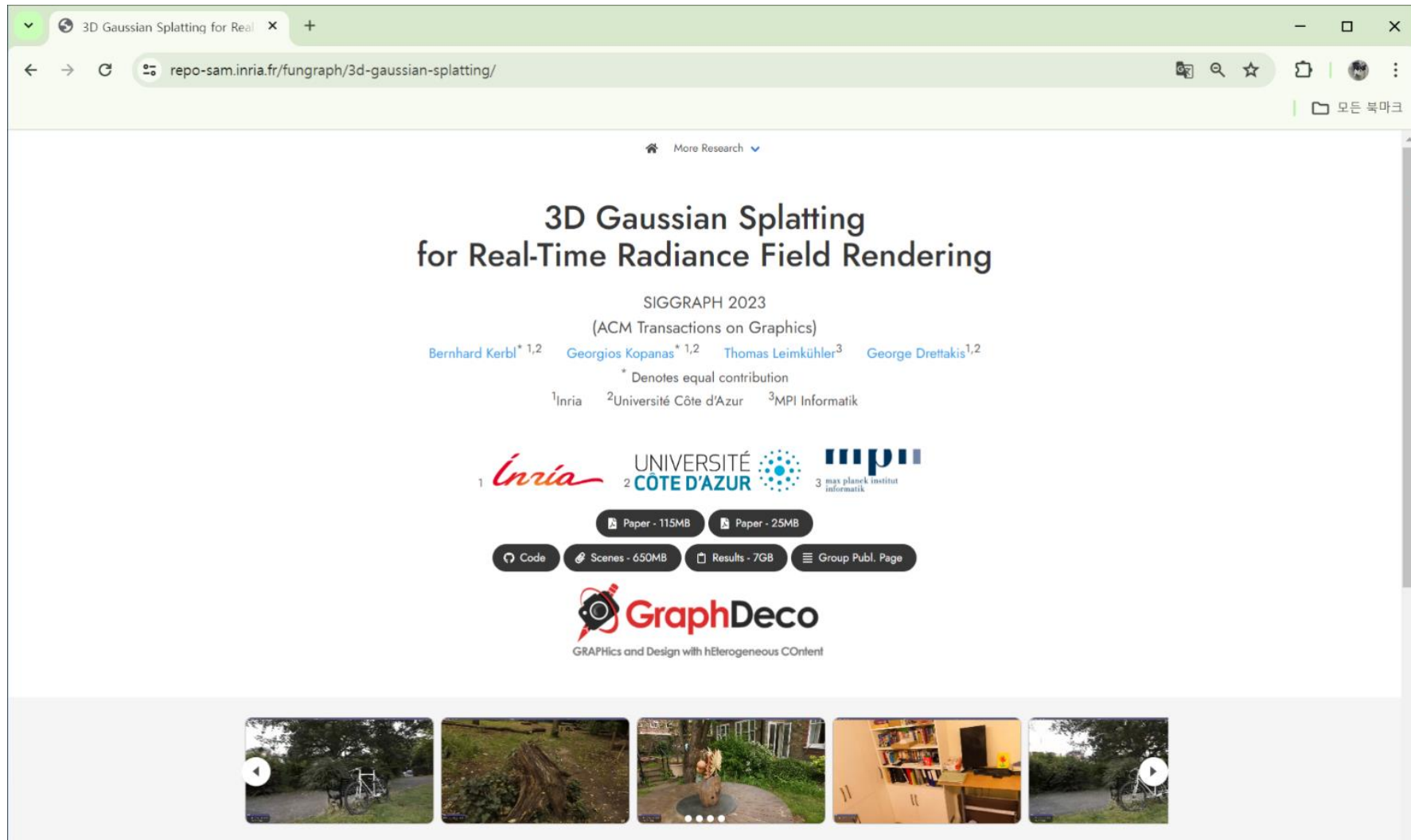
1

3D Gaussian Splatting (3DGS)

- 소개
- 예제

[1] 3DGS

- 3D 구현하는 기술중 하나
- NeRF보다 더 나은 연산속도



[1] 3DGS 예제 - 사진준비



000001.jpg



000001.png



000002.jpg



000003.jpg



000004.jpg



000009.jpg



000009.png



000010.jpg



000011.jpg



000012.jpg



000017.jpg



000017.png



000018.jpg



000019.jpg



000020.jpg



000025.jpg



000025.png



000026.jpg



000027.jpg



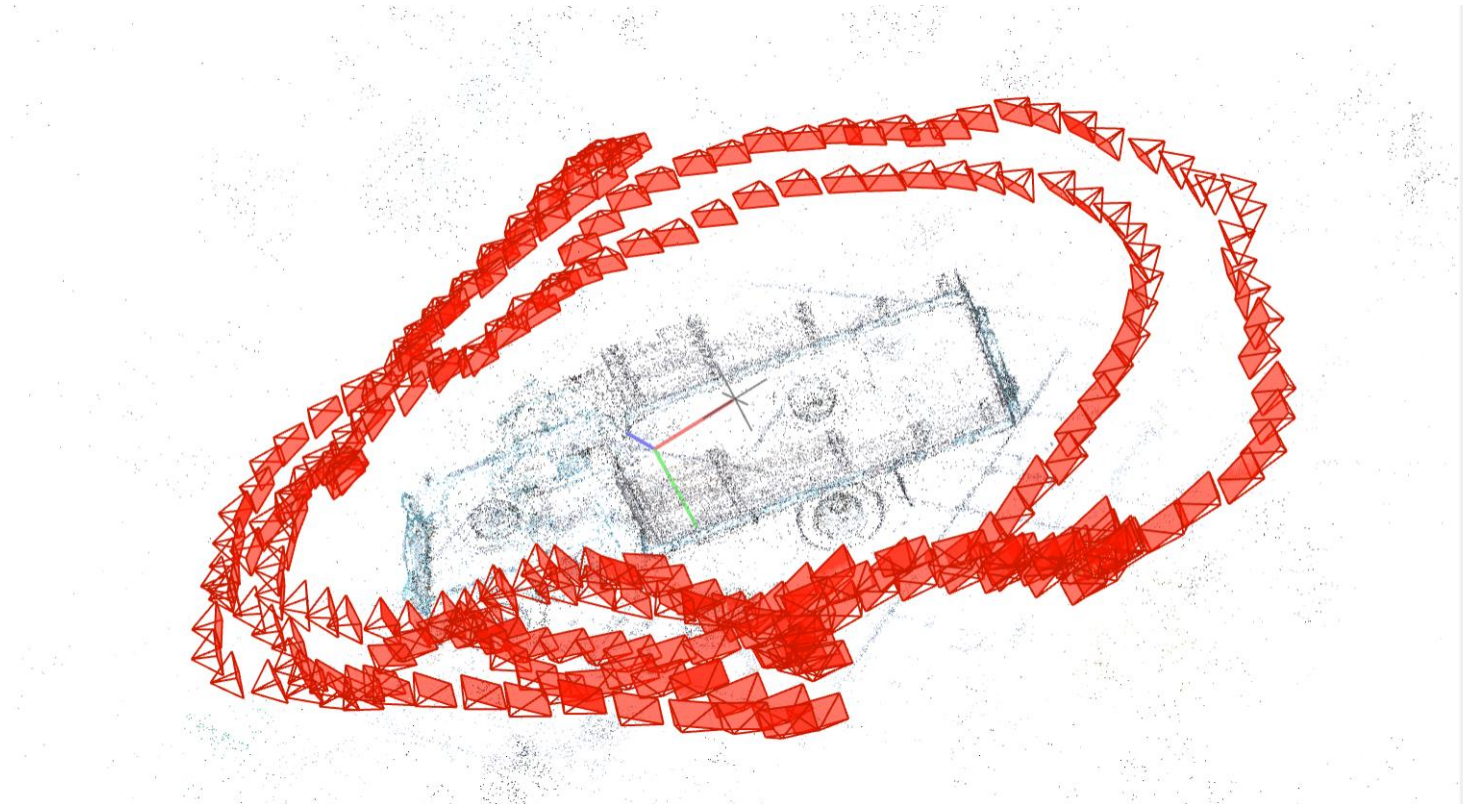
000028.jpg

[1] 3DGS 예제 - 사진준비

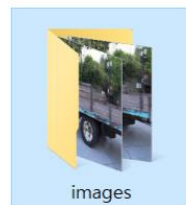


[1] 3DGS 예제 - SfM point

- 사진들의 특징으로 사진의 특징들은 매칭하기



distorted



images



input



sparse



stereo



run-colmap-geom
etric.sh



run-colmap-photo
metric.sh

[1] 3DGS 예제 - 3D Gaussian 렌더링

- 소요시간 30분

```
(gaussian_splatting) PS E:\dong\3dgs\gaussian-splatting> python .\train.py -s .\data\truck\ -r 1
Optimizing
Output folder: ./output/ac0f5dc0-1 [30/04 11:37:15]
Tensorboard not available: not logging progress [30/04 11:37:15]
Reading camera 283/283 [30/04 11:37:16]
Loading Training Cameras [30/04 11:37:16]
Loading Test Cameras [30/04 11:37:29]
Number of points at initialisation : 73119 [30/04 11:37:29]
Training progress: 23%|██████████| 7000/30000 [04:59<20:43, 18.49it/s, Loss=0.0435626]
[ITER 7000] Evaluating train: L1 0.0352414496243 PSNR 24.834679412841798 [30/04 11:42:29]

[ITER 7000] Saving Gaussians [30/04 11:42:29]
Training progress: 100%|████████████████████████████████████████████████████████████████████████████████| 30000/30000 [28:50<00:00, 17.34it/s, Loss=0.0310485]

[ITER 30000] Evaluating train: L1 0.023415937274694446 PSNR 27.570385360717776 [30/04 12:06:19]

[ITER 30000] Saving Gaussians [30/04 12:06:19]

Training complete. [30/04 12:06:33]
(gaussian_splatting) PS E:\dong\3dgs\gaussian-splatting>
```

명령어구성

```
(python .\Wtrain.py -s) (.WdataWtruckW) (-r 1)
```

(렌더링파일 실행) (₩해당이미지폴더₩) (해상도지정)

[1] 3DGS 예제 - View





▼ Metrics

59.93 (16.69 ms)

▼ 3D Gaussians

Splats

▼ Render Mode

1.000

Scaling Modifier

☒ Fast culling☐ Crop Box

▼ Camera Point view

Trackball

▼ Mode

Load camera

Save camera (bin)

Snap to closest

0

-

+

Snap to

☒

Show trackball

50.482288

-

+

Fov Y

0.009000

-

+

Near

1100.000000

-

+

Far

Key cameras: 0

Add key

Save key cameras...

Play

Play (No Interp)

Record

Stop

1.000000

-

+

Speed

Load path

Save path

☐ Save video (from playing)☐ Save frames (from playing)

Acceleration

0.300000

-

+

Speed

1.000000

-

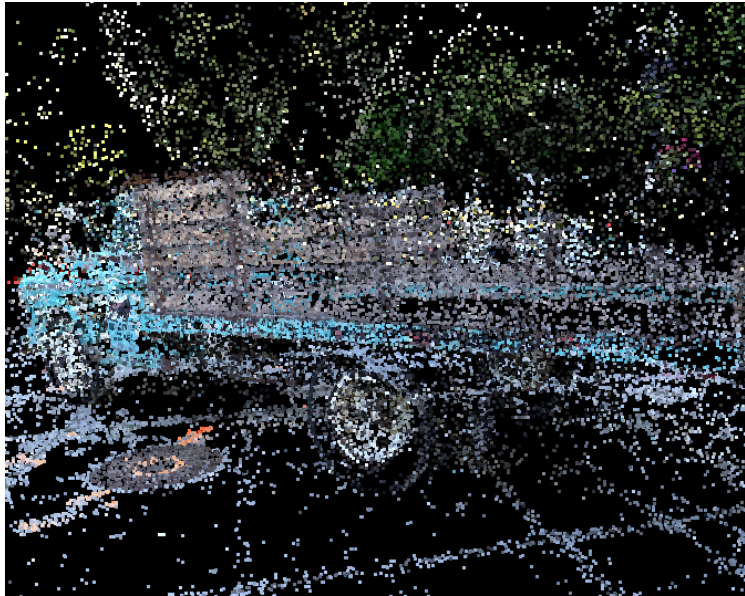
+

Rot. speed

[1] 3DGS 예제 - View

- 동작원리설명

point



Ellipsoids



split





2

3DGS실습

- 촬영환경, 방법
- 3D 구현

[2] 3DGS 실습 - 촬영환경

Insta360 PRO 2

재원

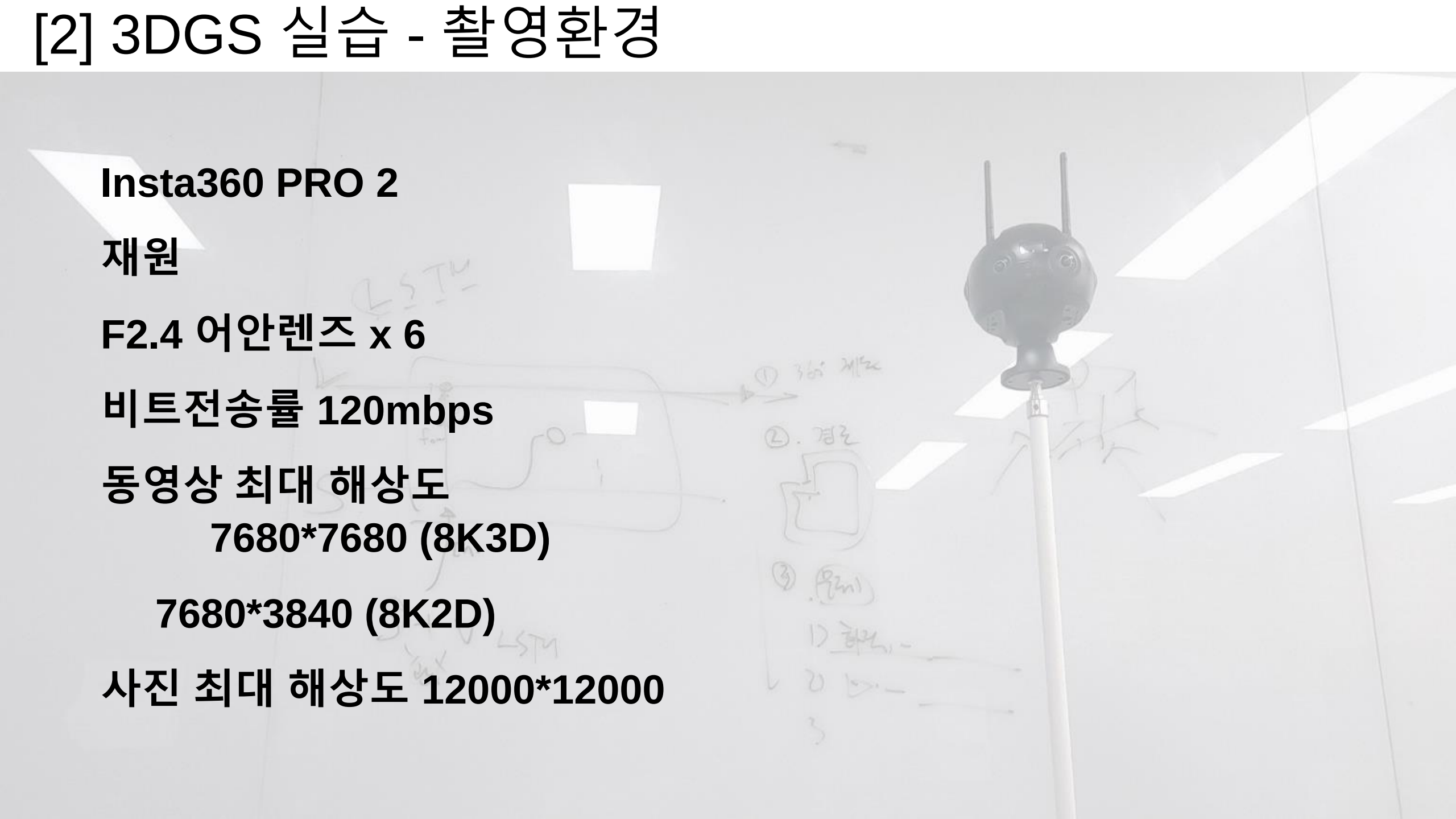
F2.4 어안렌즈 x 6

비트전송률 120mbps

동영상 최대 해상도
7680*7680 (8K3D)

7680*3840 (8K2D)

사진 최대 해상도 12000*12000

- 
- ① 360° 촬영
② 경로
③ (90m)
1) 촬영 -
2) > -
3

[2] 3DGS 실습 - 촬영방식

7680

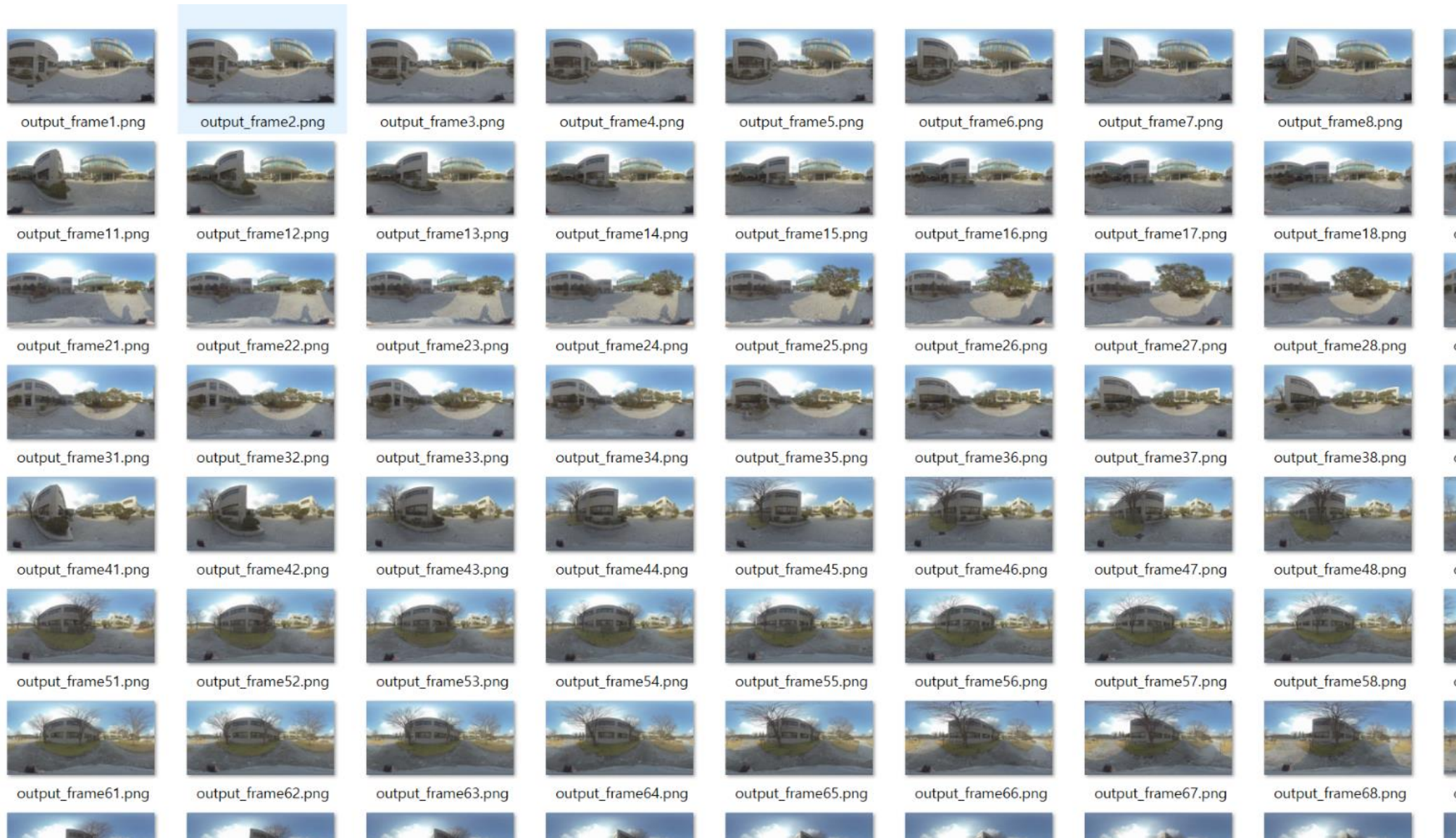
카메라높이 약
1회 1.3m 2회 2.0m 3회 2.5m
(삼각대기준)총3회 촬영

걸음속도 1m/s = 3.6km/h (초당 2걸음)
건물과 카메라간 거리: 5m
건물높이10m

3840



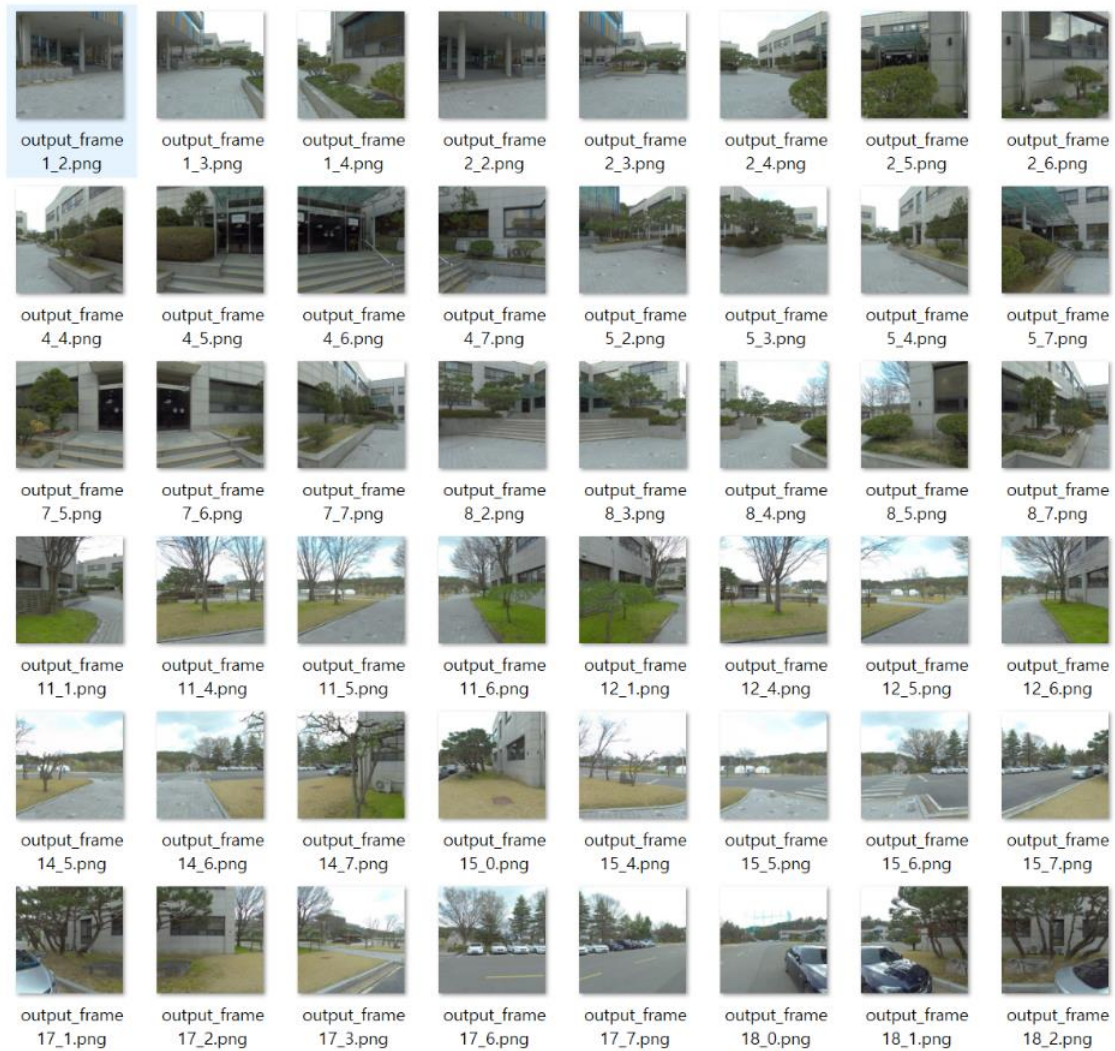
[2] 3DGS 실습 - 영상캡처



8K2D(7680*3840)



[2] 3DGS 실습 - 사진평면화



[2] 3DGS 실습 - 매핑

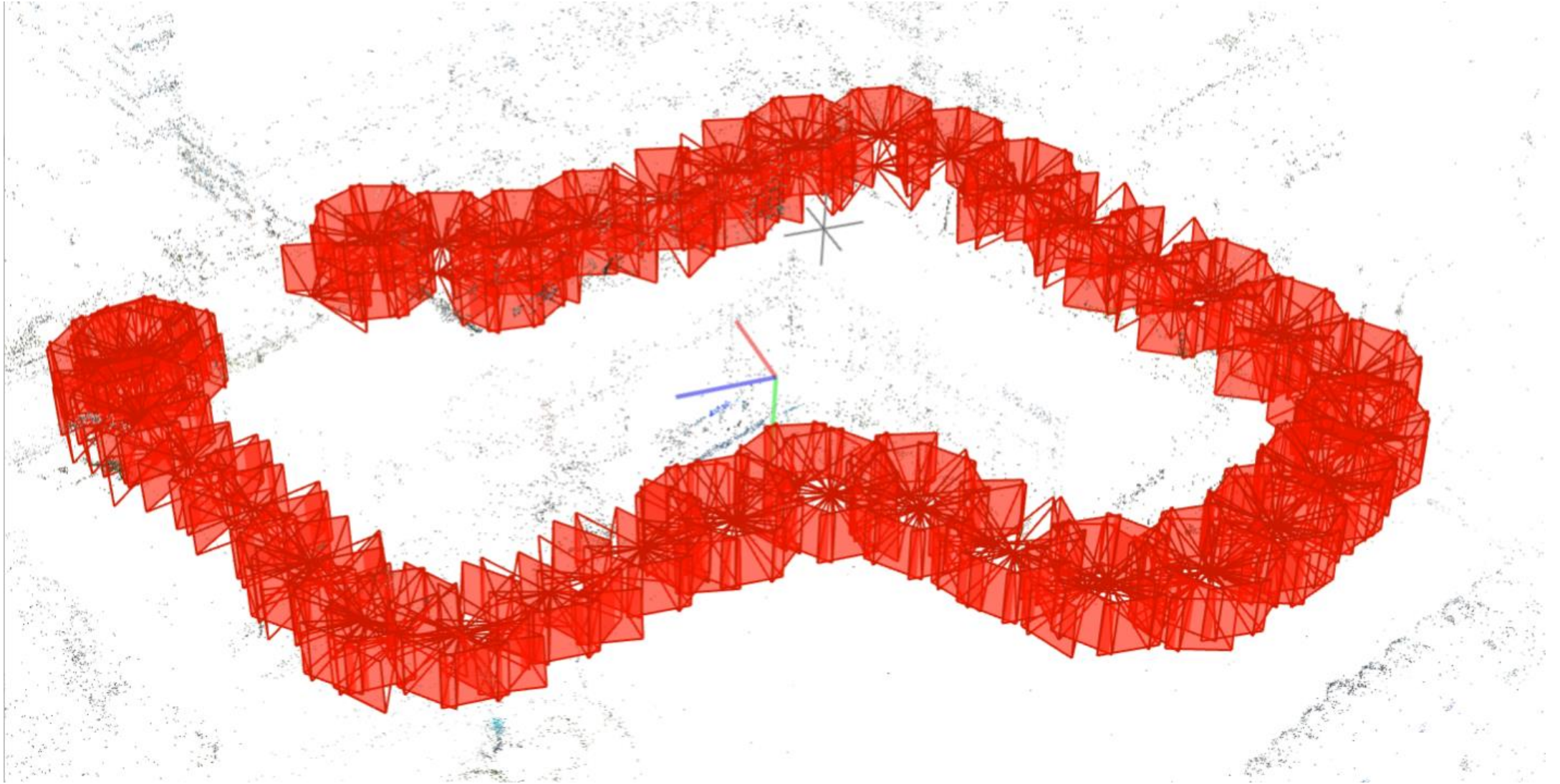
- 소요시간 30분

```
=====
Global bundle adjustment
=====

iter      cost      cost_change  |gradient|  |step|  tr_ratio  tr_radius  ls_iter  iter_time  total_time
  0  9.652029e+05   0.00e+00   2.86e+06   0.00e+00  0.00e+00  1.00e+04      0    4.45e-01    5.59e+00

CHOLMOD version 3.0.14, Oct 22, 2019: Symbolic Analysis: status: OK
Architecture: Microsoft Windows
  sizeof(int):      4
  sizeof(SuiteSparse_long):  8
  sizeof(void *):    8
  sizeof(double):    8
  sizeof(Int):      4 (CHOLMOD's basic integer)
  sizeof(BLAS_INT): 4 (integer used in the BLAS)
Results from most recent analysis:
  Cholesky flop count: 5.2193e+08
  Nonzeros in L:      7.1678e+05
memory blocks in use:      11
memory in use (MB):      0.1
peak memory usage (MB):   2.0
maxrank:      update/downdate rank:  8
supernodal control: 1 40 (supernodal if flops/lnz >= 40)
nmethods:      number of ordering methods to try: 1
  method 0: natural
    flop count: 5.2193e+08
    nnz(L):      7.1678e+05
OK
  1  9.589129e+05   6.29e+03   1.85e+04   3.61e-01   1.00e+00  3.00e+04      1    4.19e+00    9.78e+00
  2  9.589087e+05   4.22e+00   1.23e+03   5.48e-02   1.00e+00  9.00e+04      1    1.57e+00    1.14e+01
```

[2] 3DGS 실습 - SfM point



[2] 3DGS 실습 - View



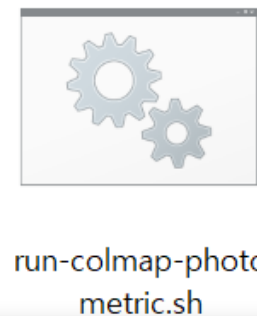
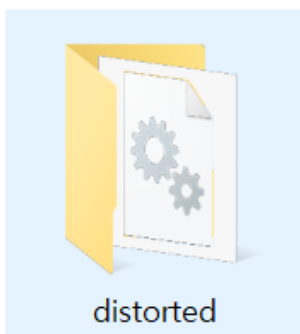


3

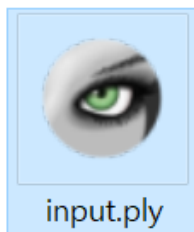
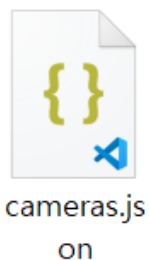
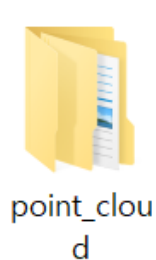
언리얼엔진5(UE5) 활용한 3DGS

- 사용배경
- 구현모습

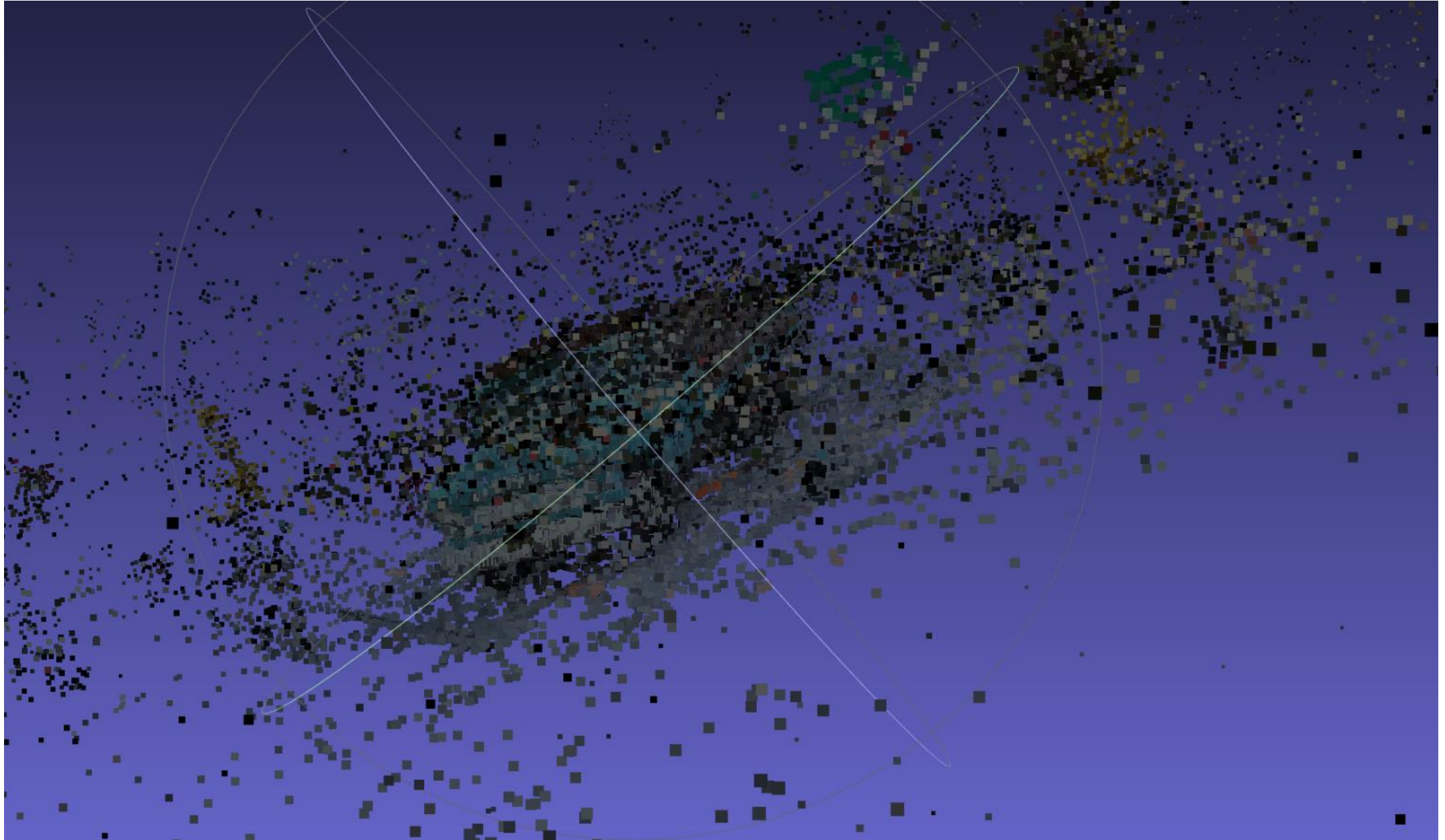
[3] UE5 활용한 3DGS - 사용배경



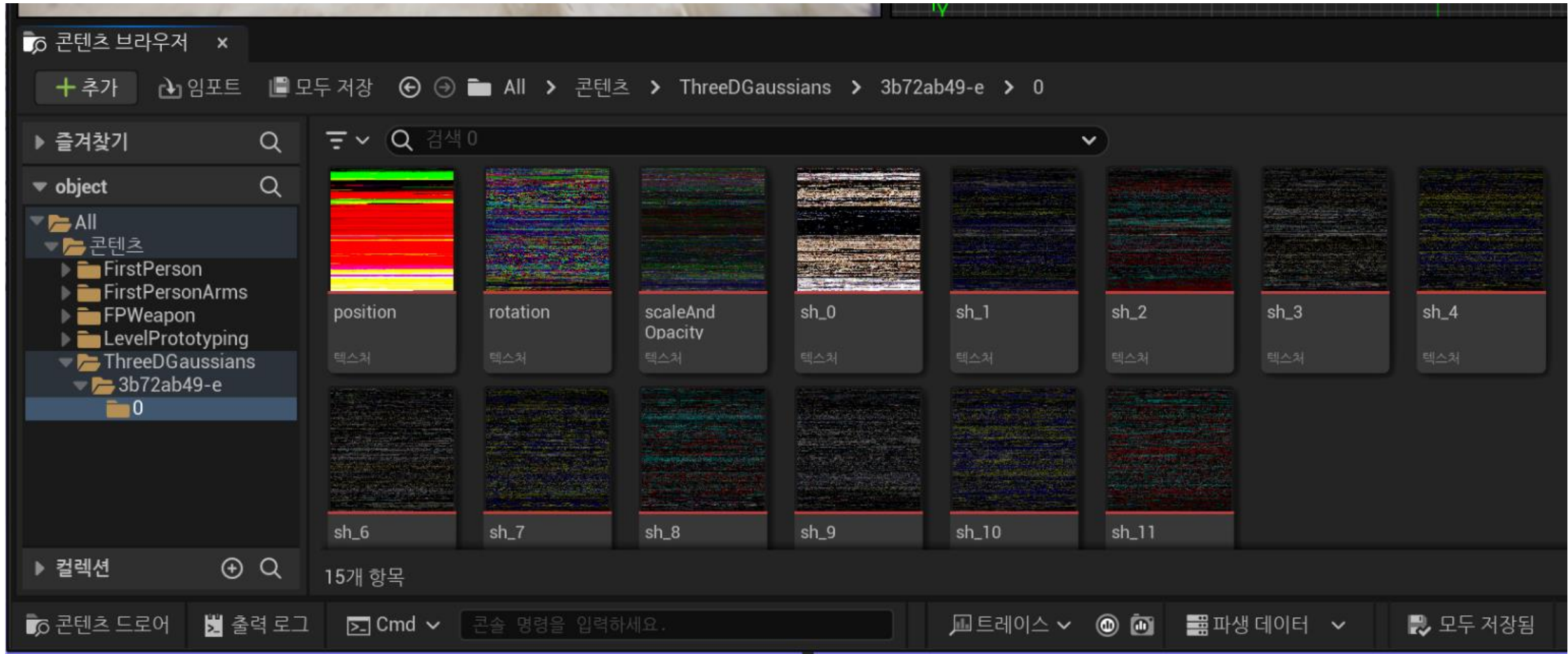
A (E:) > dong > 3dgs > gaussian-splatting > output > ac0f5dc0-1



[3] UE5활용한 3DGS - 사용배경



[3] UE5 활용한 3DGS - 플러그인



[3] UE5 활용한 3DGS - 구현



[4] 문제해결

- 실제 적용예의 문제점

이미지손실

평면화과정 이미지 손실문제
중앙으로 객체를 포커싱



촬영설정 문제점

일정한 색감의 영상을 받지 못한 문제
촬영 색영역 설정 후 재촬영



대용량 파일

중복도 최소화(최소권장3개)

크기:	11.1GB (12,017,753,200 바이트)
디스크 할 다 크기.	11.1GB (12,025,499,648 바이트)

크기:	6.04GB (6,488,877,412 바이트)
디스크 할 다 크기.	6.04GB (6,492,762,112 바이트)

4

3DGS실습간 발생한 문제

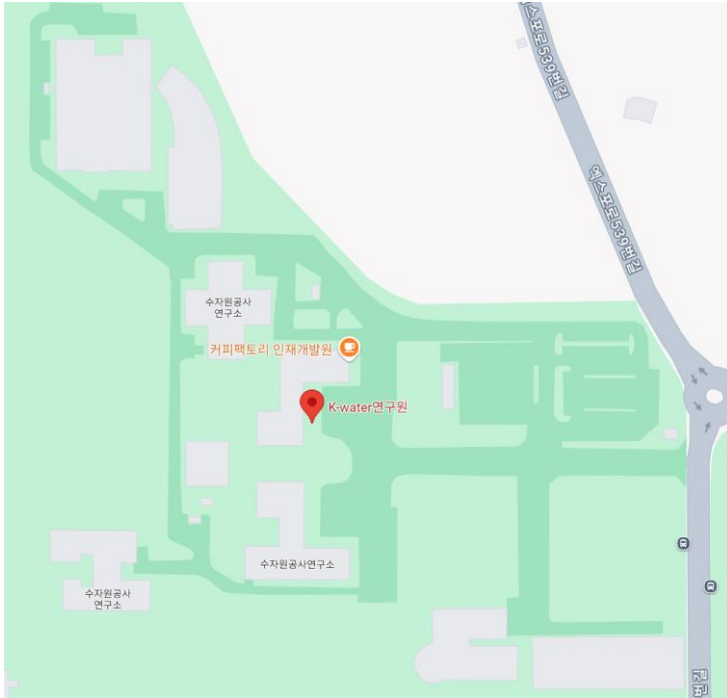
- 문제점과 해결방안
- 차후계획



[4] 차후 계획

본관

건물 크기의 객체가능성 확인



정수장

수자원 주요 시설물 촬영및 구현



공사현장

계속 바뀌는 현장에 대한 안전및
진행도확인

